

**GRAPH THEORY EXERCISES
DISCRETE MATHEMATICS**



Lecturer :

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INFORMATIC MANAGEMENT MAJOR

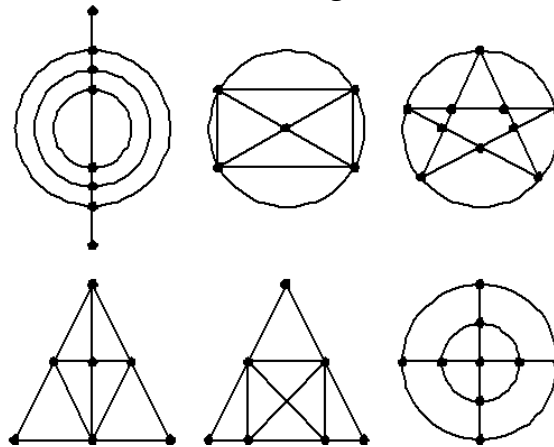
FACULTY OF VOCATION

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The Exercises

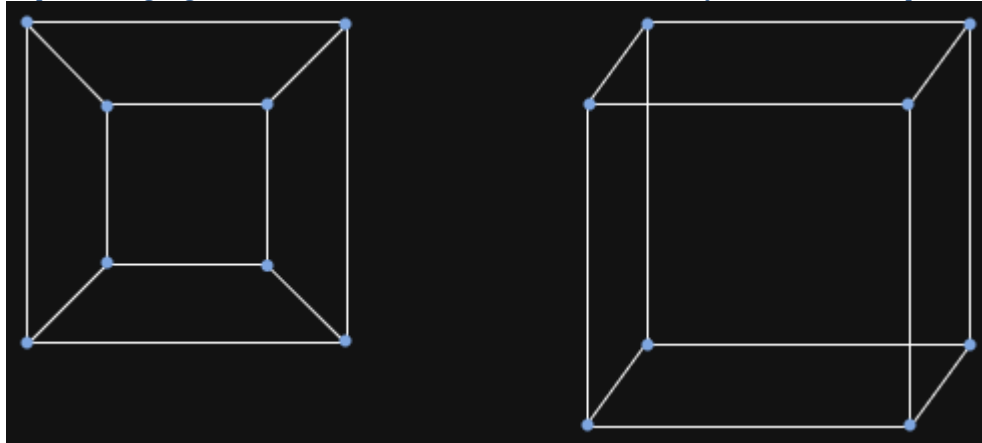
1. Draw two graphs that are isomorphic to a regular graph of degree 3 that has 8 vertices.
2. Does K_{13} have an Euler circuit? A Hamilton circuit? Repeat the same question for K_{14} .
3. Which of the graphs below can be drawn in a single stroke without lifting the pencil at all?



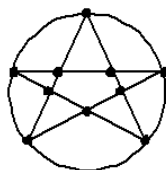
The Answers

1. Two graphs isomorphic (using draw.io)

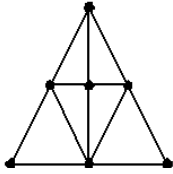
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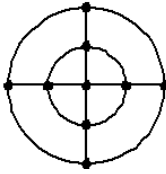
2. In graph theory, an Euler circuit exists if all vertices in the graph have even degrees. A Hamilton circuit exists if the graph contains a cycle that visits every vertex exactly once. For the complete graph K_{13} , each vertex has degree 12, which is even. Therefore, K_{13} has an Euler circuit. Since all complete graphs with at least 3 vertices are Hamiltonian, K_{13} also has a Hamilton circuit. For the complete graph K_{14} , each vertex has degree 13, which is odd. Therefore, K_{14} does not have an Euler circuit. However, K_{14} still has a Hamilton circuit because it is a complete graph.
3. A graph can be drawn without lifting the pencil if it has an Euler path or Euler circuit. This occurs when the graph has either zero or exactly two vertices with odd degrees. If there are more than two vertices with odd degrees, then it is not possible to draw the graph in one stroke.



This graph can be drawn in one stroke because all vertices have even degrees, so it contains an Euler circuit.



This graph can be drawn in one stroke because it has exactly two vertices with odd degrees, so it contains an Euler path.



This graph can be drawn in one stroke because all vertices have even degrees, so it contains an Euler circuit.